

Roll No.

Total No. of Pages : 02

Total No. of Questions : 16

B.Sc. (IT/Graphics & Web Designing)/BCA (Sem.-3)

DATA STRUCTURES

Subject Code : UGCA-1915

M.Code : 78181

Time : 3 Hrs.

Max. Marks : 60

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains SIX questions carrying TEN marks each and students have to attempt any FOUR questions.

SECTION-A

Write briefly :

- 1) Garbage collection
- 2) Linked List
- 3) Small 'o' Notation
- 4) Define Queue
- 5) Polish Notation
- 6) Define Heap
- 7) In-degree
- 8) AVL Tree
- 9) Searching
- 10) Binary Tree

SECTION-B

- 11) Give an example of postfix expression. How to evaluate it using stacks?
- 12) Differentiate between arrays and linked list.
- 13) Explain the working of quick sort with example.
- 14) Define hashing and hash table. Also explain the concept of collision and its resolution.
- 15) Write a note on AVL tree and B-Trees.
- 16) Write algorithms for creating a stack using linked list. How to add and delete a note from stack?

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SECTION-B

2. Write the algorithm for selection sort. Explain it with the help of an example.
3. Write algorithm for breadth first search and explain it with the help of an example.
4. Define data structure. Explain different types of data structures.
5. Write algorithm for linear search. Give an example when the element is not found in the list.
6. Write short note on :
 - a. Doubly linked list
 - b. B Tree
7. What is a binary tree? Explain basic tree terminologies used for binary tree.

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